

Soumen Roy

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Experience

Centre for Cosmology, Particle Physics and Phenomenology (CP3), UCLouvain, Postdoctoral Researcher Louvain-la-Neuve, Belgium
2024 – present
2 years

- Active member of the LIGO–Virgo–KAGRA Collaboration.

Nikhef / Utrecht University, Postdoctoral Researcher The Netherlands
2020 – 2024
4 years

Education

Ph.D. Indian Institute of Technology Gandhinagar, Physics India
2014 – 2021

- Thesis: Aspects of gravitational wave data analysis from compact binary coalescences
- Supervisor: Prof. Anand S. Sengupta

M.Sc. The University of Calcutta, Physics (Astrophysics & Astroparticle Physics) India
2011 – 2013

- Thesis: An Introduction to Fluid Dynamics and Numerical Solution of Shock Tube Problem by Using Roe Solver
- Supervisor: Prof. Sanjay Kumar Ghosh

B.Sc. Vidyasagar University, Physics India
(Honours) 2008 – 2011

Computational Expertise

Programming and workflow: Python, C/C++, HTCondor, bash, LaTeX; experience with MATLAB, Mathematica, Gnuplot, and FORTRAN

Gravitational-wave software: LALSuite, PyCBC, Bilby, BayesWave; Bayesian inference and waveform-model implementation

Computing environments: Linux, macOS, Windows; high-throughput and high-performance computing for large-scale gravitational-wave analyses

Mentoring

- 2025–Present · Jef Heynen (CP3, UCLouvain), graduate:** Leveraging the null stream to detect strongly lensed gravitational wave signals (with Justin Janquart)
- 2025–Present · Ioannis Liodis (DESY), graduate:** Impact of astrophysical environment surrounding compact binaries on cosmology inference (with Samaya Nissanke)
- 2022–2024 · Giada Caneva Santoro (IFAE, UAB), graduate:** Measuring the astrophysical environment surrounding compact binaries with second-generation and future detectors (with Mario Martinez)
- 2022–2025 · Abhishek Sharma (IIT Gandhinagar), graduate:** Designing a search framework for detecting signals beyond general relativity; developing a rapid parameter estimation framework (with Anand S. Sengupta)
- 2021–2022 · Anna Puecher (Nikhef / Utrecht University), graduate:** Testing general relativity using higher-order modes of gravitational waves from binary black holes (with Chris Van Den Broeck)
- 2025 · Ioannis Liodis (GRAPPA, University of Amsterdam), master's:** Impact of astrophysical environment surrounding compact binaries on cosmology inference (with Samaya Nissanke)
- 2022–2023 · Bram van de Zande (Utrecht University), master's:** Black holes, or black hole mimickers (superspinars)? Gravitational waves as a means to tell the difference (with Chris Van Den Broeck)
- 2021 · Marien Hofstee (Utrecht University), bachelor's:** Higher harmonics in gravitational wave signals from binary black hole mergers (with Chris Van Den Broeck & Fabian Ziltener)
- 2019–2021 · Harsh Narola (IIT Gandhinagar), master's:** Beyond general relativity: designing a template-based search for exotic gravitational wave signals (with Anand S. Sengupta)
- 2019 · Mohini Umesh Ramwala (IIT Gandhinagar), master's:** GR and non-GR template banks for LIGO data analysis (with Anand S. Sengupta)

Teaching

- Spring 2017:** Graduate Teaching Fellow, IIT Gandhinagar: Statistical Mechanics
- Fall 2018:** Graduate Teaching Fellow, IIT Gandhinagar: Computational Physics
- Spring 2019:** Graduate Teaching Fellow, IIT Gandhinagar: Introduction to Computing

Selected Talks and Presentations

- *Line-of-sight acceleration in compact binaries: the role of higher harmonics and eccentricity* — XVI Einstein Telescope Symposium, Aachen, Germany (June 2026)
- *Scalar fields around black hole binaries in LIGO-Virgo-KAGRA* — New opportunities to detect gravitational waves and dark matter in orbital dynamics, Benasque Science Center, Spain (December 2025)
- *Fast likelihood evaluation of eccentric-precessing binaries using relative binning* — 4th Einstein Telescope Annual Meeting, Opatija, Croatia (December 2025)
- *Testing modified gravity with the eccentric neutron star-black hole merger GW200105* — Belgian-Dutch Gravitational Wave Meeting, Radboud University Nijmegen, Netherlands (2025)
- *Testing modified gravity with the eccentric neutron star-black hole merger GW200105* — Joint Theoretical Gravitational Wave Seminars, Leuven, Belgium (2025)
- *Parametrized test of general relativity with gravitational waves: including higher harmonics and precession* — Belgian Physical Society general scientific meeting, Louvain-la-Neuve, Belgium (2025)
- *Probing gravitational waves using GNSS constellations* — Tabletop Scale Cosmology, online conference (2025)
- *Probing gravitational waves using GNSS constellations* — 59th Rencontres de Moriond on Gravitation, La Thuile, Italy (2025)
- *Can compact binary coalescences in environments pose as in vacuum?* — biweekly BelGrav meeting, online (December 2024)
- *Update on TIGER analysis with GW230814* — LIGO-Virgo-KAGRA Collaboration Meeting, Barcelona, Spain (September 2024)
- *A non-orthogonal wavelet transformation for reconstructing gravitational wave signals* — APS April Meeting, New York (April 2022)
- *Detection of non-quadrupole modes of gravitational waves from multiple inspiralling black holes* — 9th International Conference on Gravitation and Cosmology, IISER Mohali, India (December 2019)
- *Effectual template banks for upcoming compact binary searches in Advanced-LIGO Virgo data* — 30th meeting of the Indian Association for General Relativity and Gravitation, BITS Pilani, Hyderabad, India (January 2019)
- *Designing an effectual template placement algorithm for searches of gravitational wave from compact binary coalescences* — Exploring the Universe: Near Earth Space Science to Extra-Galactic Astronomy, SNBNCBS Kolkata, India (November 2018)
- *Designing an effectual template placement algorithm for gravitational wave searches* — Summer School: Gravitational Waves 2018, École de Physique des Houches, France (July 2018)
- *Hybrid geometric-random template placement algorithm for gravitational wave searches from compact binary coalescences* — 29th meeting of the Indian Association for General Relativity and Gravitation, IIT Guwahati, India (May 2017)
- *A hybrid geometric-random template placement algorithm for gravitational wave searches from compact binary coalescences* — Time Series Analysis for Synoptic Surveys and Gravitational Wave Astronomy, ICTS Bangalore, India (March 2017)

Professional Memberships

Since 2020: Member of Einstein Telescope Observational Science Board

Since 2020: Member of Virgo Scientific Collaboration

Since 2017: Member of LIGO Scientific Collaboration

2017–2020: Member of LIGO-India Scientific Collaboration

Journal Referee

- Physical Review D, Physical Review Letters